

2

What is consciousness?

→ Mark Sprevak

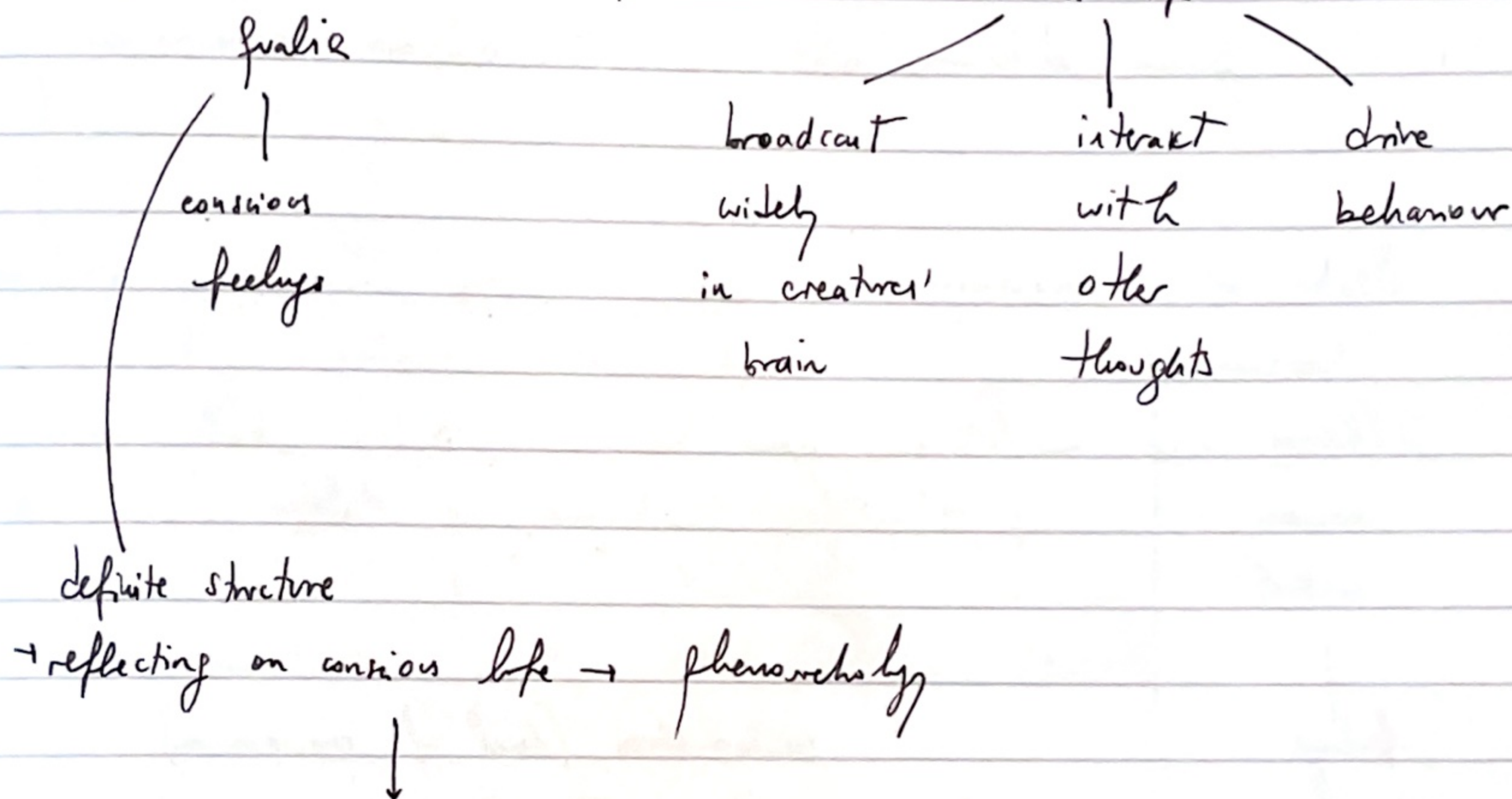
What do we mean when we say something / someone is conscious

— smart, aware

— awake — global state

— access consciousness (Ned Block) — a thought is access

— phenomenal consciousness



Phenomenology

GAP

Brain Science

Philosophers → Frank Jackson: thought experiment → Mary and red colour

Solving this issue means showing how every aspect of phenomenal consciousness is determined by brain activity

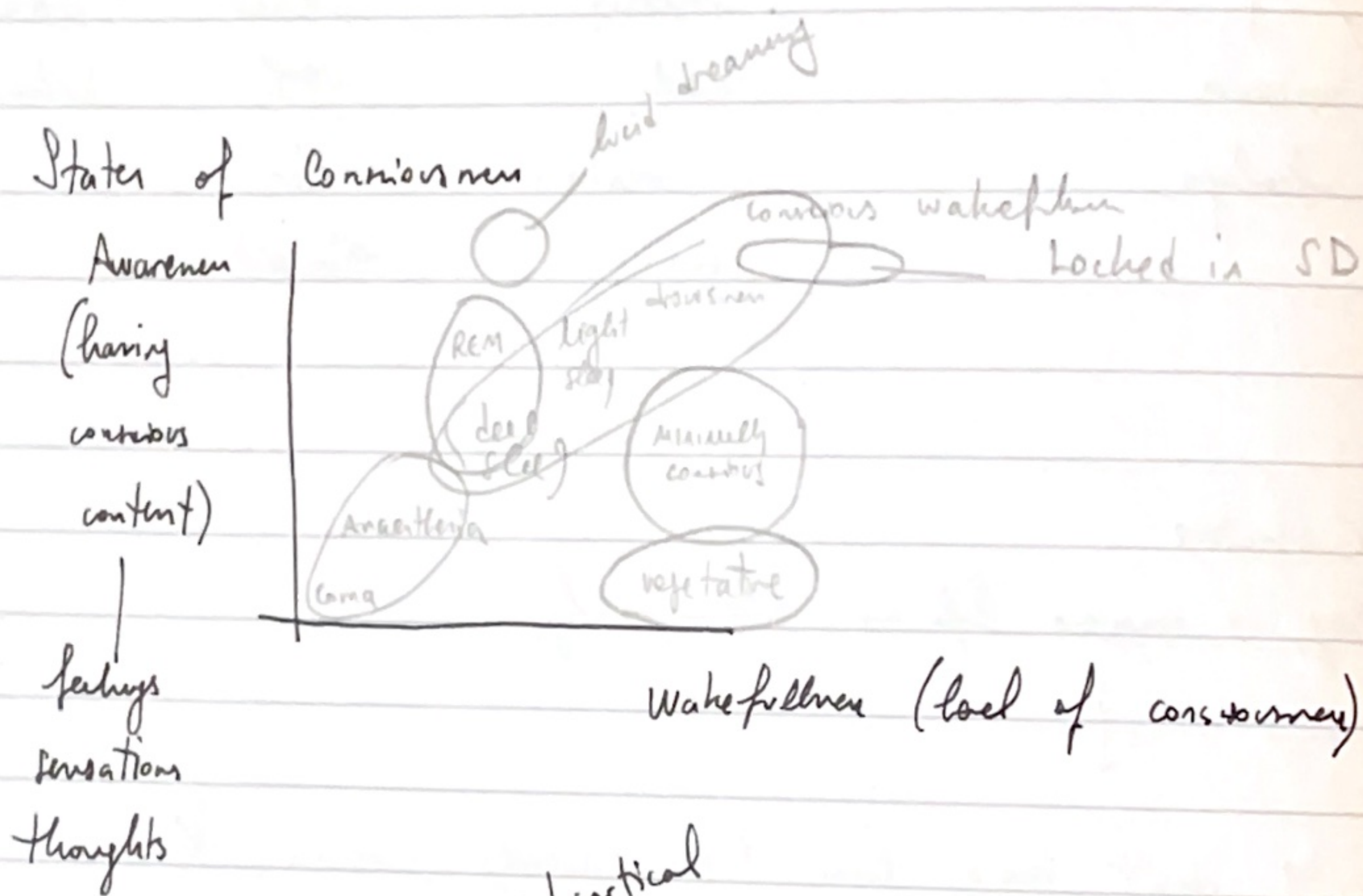
2.2

→ David Cornell

→ Current questions today fall under NCC (Neural Correlates of Consciousness)

1) What neural activity determines the state or level of consciousness that a person is in.

2) What governs determine the content of our consciousness



⦿ Reticular formation + thalamus ↔ Wakefulness
Cortex ↔ Awareness

Awareness — ext → frontal + parietal lobes
internal — medial side

} Activity is negatively correlated (when one is ↑ the other is low)

As we fall asleep → areas in cortex ← but changes in wakefulness affect cortex
communicate ↓ with each other

→ We are aware of a small subset of things in our environment

Inattention blindness

⇒ Attention set → what we're looking for

Bistable images

≠ interpretation

only one at a time

perceptual switches

dissociation → perception vs. awareness

researchers look which area of brain would be active at the same time that as perceptual switches

correlation with occipital, parietal, frontal lobe

Studying causality → TMS (transcranial magnetic stimulation)

applies a brief powerful magnetic pulse to the surface of the head

↓ interferes temporarily

activity of the area of cortex underneath it

TMS applied to bistable images ⇒ different parts of parietal cortex play a different role

UNCONSCIOUS
PERCEPTION

→ experiment → showing people things but actively ignoring them

↓ Backward visual masking

switches slower faster

1st time — marked graving

↓
correlated images with words → semantically related

↓
the marked word had
activated a semantic
network in the brain